Pool AI — Current Architecture (Phase 7 / 8)

Token-based transformer actor-critic for 14.1 Continuous pool. 566,918 parameters. Trained on CPU.

Design philosophy

Pool shot-making has three layers: **geometric** (which balls can be pocketed into which pockets; line-of-sight checks; ghost-ball aim), **strategic** (which of the makeable shots to take, for position on the next shot), and **mechanical** (force and spin for cue ball control). We handle each layer with the right tool:

- **Geometric layer**: deterministic algorithm enumerates legal (ball, pocket) pairs. No learning needed.
- **Strategic + mechanical layers**: learned by the network — picking among legal shots and choosing force/spin.

This is the AlphaZero pattern adapted to pool: the solved parts are computed exactly; the parts that need judgment are learned.

Decision pipeline (per shot)

1. Geometric enumeration (Python, no learning). For the current cue and ball positions, enumerate all (ball, pocket) pairs where:

- The **ghost-ball position** (where the cue ball's center must be at contact to send the object ball toward the pocket) is on the table — not behind a cushion.
- The **cue** → **ghost** path is clear of other balls (line-of-sight with 2R + 0.3" clearance).
- The **ball** → **pocket** path is clear of other balls.
- The **cut angle** is ≤ 75° (very steep cuts discarded).
- The cue is behind the ball relative to the pocket direction.

Output: a list of *LegalShot* records, each with ball_id, pocket_idx, ghost position, aim angle, cut angle, and distances.

2. Network inference on token batch:

The board state + legal shots become a token sequence fed to a transformer. Each token type has features:

Token type	Count	Features
Ball	up to 16	(x, y, is_cue)
Pocket	6	(x, y, is_corner)
Legal shot	up to 60	(ghost_x, ghost_y, target_ball_x, target_ball_y, target_pocket_x, target_pocket_y, cut_angle_deg/90, cue→gh

3. Action sampling:

- **Shot index**: categorical over legal shots (softmax of per-shot score logits).
- **Force**: Gaussian sample around the chosen shot's force_mean; sigmoid to [50, 250] in/s.
- **Spin**: Gaussian sample around chosen shot's spin_mean; tanh to [-2, +2] (spin factor relative to natural roll).
- **Aim**: derived geometrically from the chosen shot's ghost — no learning needed.

Network: PoolGameNet

Total parameters: **566,918**. Implementation in rl/pool_game_net.py.

Encoders

```
ball_encoder:   Linear(3 → 128) → LayerNorm → GELU
pocket_encoder: Linear(3 → 128) → LayerNorm → GELU
shot_encoder:   Linear(9 → 128) → LayerNorm → GELU
type_embed:     Embedding(3 types → 128), added to each token
```

Transformer

```
TransformerEncoder(
    layers        = 4,
    embed_dim     = 128,
    num_heads     = 8,
    ff_dim        = 256,    # 2 × embed_dim
    activation    = GELU,
    norm_first    = True,   # pre-LN for stable training
)
# Padding mask hides invalid ball/shot slots from attention.
```

Output heads (per legal shot)

```
shot_head: Linear(128 → 128) → GELU → Linear(128 → 3)
           # outputs (score_logit, force_mean, spin_mean) per shot
           # softmax over all valid shots → which shot to take
log_std:   nn.Parameter of shape (2,) for (force_std, spin_std)
           # clamped at log_std_min = -2.5
```

Value head (state value)

```
value_head: Linear(128 → 128) → GELU → Linear(128 → 1)
            # input is mean-pool over ball + pocket tokens
            # shot tokens excluded from the pooling
```

Training

Standard PPO with joint categorical-continuous action space:

```
log_prob = log_softmax(scores)[shot_idx]
         + Normal(force_mean[shot_idx], force_std).log_prob(force_raw)
         + Normal(spin_mean[shot_idx],  spin_std).log_prob(spin_raw)

entropy  = Categorical(scores).entropy()
         + Normal(0, force_std).entropy()
         + Normal(0, spin_std).entropy()

loss = pg_loss + 0.5 · v_loss − 0.01 · entropy
```

Default hyperparameters

Hyperparameter	Phase 7 value	Phase 8 value
Optimizer	Adam	Adam
Learning rate	1e-4	3e-5 (gentler, avoid catastrophic forgetting)
Env parallelism	16	16
Steps per update	32	32
PPO epochs / clip	4 / 0.2	4 / 0.2
Value coef	0.5	0.5

Entropy coef	0.01	0.01
log_std min	-2.5	-2.5
Iters	500–1500	500

Environments

- **Phase 7 (Phase7Env)**: full 15-ball rack; opening break auto-executed in reset(); agent takes over from shot 2. Includes rerack mechanic.
- **Phase 8 (Phase8Env)**: break-ball drill. Rack is 14 balls (apex empty) + break ball placed in classic 14.1 positions + cue in kitchen. Agent's first shot is the break. If break succeeds, continues as full run-out (self-balancing training distribution).

Reward structure

- **+10** per object ball pocketed on a shot (counts incidentals when the called shot succeeds).
- **-10** scratch penalty (cue pocketed).
- Episode ends on miss (called ball not in called pocket), scratch, or max_shots.
- The opening break is exempt from strict call-shot — any pocketed ball counts, matching real 14.1 break rules.

Legal-shot enumeration (critical detail)

This is the algorithmic heart of the system. Done right, it ensures the network only ever picks from physically realizable shots.

Ghost ball geometry

```
ghost = ball − 2R · (pocket − ball) / |pocket − ball|
# 2R-offset from ball, away from the pocket direction.
# This is where the cue ball center must be at the moment of contact
# to send the object ball straight toward the pocket center.
```

Line-of-sight check

```
def segment_blocked(p1, p2, obstacles, exclude_ids, clearance=2R+0.3):
    # Parameterize segment from p1 to p2.
    for each obstacle b not in exclude_ids:
        t = projection of (b − p1) onto segment direction
        if t < 0:   continue          # obstacle is behind p1
        if t > seg_len:
            if |b − p2| < clearance: return BLOCKED   # near endpoint
        else:
            if perp(b to segment) < clearance: return BLOCKED
    return CLEAR
```

Subtle detail: obstacles behind the cue ($t < 0$) are NOT considered blockers. A ball touching the cue ball on the back side does not prevent the cue from moving forward. This was a bug in an earlier version — fixed.

Cut angle constraint

The cut angle is the angle between the cue-approach direction (toward ghost) and the intended ball-exit direction (toward pocket). Shots with cut > 75° are discarded — they require too-thin hits to be reliable.

Action space details

Component	Type	Raw range	After activation
shot_idx	categorical	{0..N_legal–1}	chosen shot from legal set
force_raw	continuous	(–∞, ∞)	sigmoid → [50, 250] in/s
spin_raw	continuous	(–∞, ∞)	tanh → [–2, +2] (relative to natural roll)
aim_angle	—	derived	atan2 toward ghost of chosen shot

Spin interpretation: spin_factor = ω_initial / (v_initial / R). 0 = pure linear (develops natural roll via sliding friction); +1 = natural roll from the start; +2 = max follow; –1 = full backspin; –2 = max draw.

What led here (brief)

- **Phases 1–6**: flat-observation transformer (*PoolAttentionNet*, 438K params). Learned aim via continuous aim output. Plateaued because implicit ball selection via aim angle + sparse reward couldn't teach strategic shot choice from scratch.
- **Key insight**: "which ball to shoot" is a solved geometric problem; only "which of the makeable shots, with what force and spin" needs learning. This is the AlphaZero insight applied to pool.
- **Phase 7**: rebuilt with token-based network and legal-shot enumeration as a pipeline stage. Network picks among enumerated shots rather than generating aim directly. Immediate jump in play quality.

- **Phase 8**: break-ball drill env. Self-balancing curriculum: if break succeeds, episode continues as a run-out, so good breaks are rewarded by cascading return, bad breaks end immediately. Warm-started from Phase 7 and fine-tuned at low lr to preserve general-play skills.

Open areas

- **Search at inference** (AlphaZero-style MCTS over shots × force/spin samples). Currently inference uses deterministic argmax + mean output. Adding tree search should boost play.
- **Scaling network capacity**: 566K may not be the bottleneck yet — we'll know by whether plateau metrics improve with 1–2M parameters. No move until warranted.
- **Specialized drills**: Phase 8 is a break-ball drill; similar drills for "save the break ball" endgame, cluster-management, and safety play would further sharpen specific strategic concepts.